Role Prompts Are Policy Controls: Role-Conditioned Fairness Evaluation for Civic AI Systems

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Abstract

Civic AI systems are commonly deployed with role prompts: assistant, advocate, mediator, news provider, government information service, campaign aide, or researcher. These roles are often treated as style settings. We argue that they are policy controls. A role prompt can change what help is allowed, when the system refuses, whether matched civic viewpoints receive symmetric treatment, and whether an answer fits the institutional duty implied by the role. We introduce a role-counterfactual evaluation for civic AI systems: hold the prompt, topic, source packet, model, and judge fixed, then vary the assigned role and measure refusal, viewpoint symmetry, role-profile fit, non-refusal quality, judge robustness, and targeted remediation. In a 2100-row baseline over five local models, seven roles, thirty civic prompts, and two role-presentation modes, the primary judge observes a 14.3% refusal rate, a 13.1% over-refusal rate, and 72 one-sided refusals among 420 mirrored-pair comparisons. An executable role-policy intervention reduces one-sided refusals to 37 pairs and lowers the full-grid refusal rate to 13.4%, but it also slightly reduces judged non-refusal quality. On a targeted 300-row failure sample, refusal falls from 81.0% to 51.0% and role-profile fit improves. The lesson is not that one role policy solves civic fairness. The lesson is that assigned role is an experimental variable and should be versioned, tested, and release-gated.

1 Introduction

Civic AI systems do not just answer questions. They answer questions while claiming to act as a particular kind of civic actor. A general assistant, campaign aide, government information service, mediator, researcher, advocate, and news provider can all receive the same civic prompt and the same source packet, but each role carries different obligations. Some roles may help the user craft a persuasive argument. Some should not persuade at all. Some should summarize procedure. Some should map disagreement. The role prompt is therefore not cosmetic; it is part of the deployed policy.

This creates a problem for fairness evaluation. A flat question such as "is this model politically biased?" misses several deployment failures that matter to users. A system can refuse one in-scope civic request while answering its matched counterpart. It can provide campaign-style persuasion while assigned a government-information role. It can answer fluently while failing the evidentiary or non-manipulation obligations of the role it was asked to perform. These are role-conditioned failures, not just differences in tone.

We propose role-conditioned fairness evaluation. The evaluation asks what changes when the assigned role changes under otherwise matched conditions. It is designed for practical governance: a team should be able to discover failures, change the role policy, rerun matched comparisons, and block a release if the new policy worsens civic safety or fairness.

Our contributions are:

1. a role-counterfactual civic evaluation design with mirrored civic prompt pairs;

2. a metric split between refusal, over-refusal, viewpoint symmetry, role-profile fit, and non-refusal quality;
3. a matched remediation experiment that treats role-policy instructions as the intervention;
4. targeted ablations, stress prompts, and a failure-regression gate for release monitoring;
5. a human-calibration protocol, not yet imported in the current results, for checking the LLM-judge conclusions on hard cases.

2 Evaluation Design

Unit of evaluation. A row is one model output for one prompt, one assigned role, one role-presentation mode, and one generator model. The main local evaluation uses five local generator models, seven civic roles, thirty prompts, and two role-presentation modes, producing 2100 judged outputs. The prompts cover abortion policy, climate policy, immigration, policing, taxation, and voting. Each prompt is paired with a static source packet so the model is not rewarded for inventing current facts.

Role counterfactuals. For a fixed model and prompt, the assigned role is varied across assistant, advocate, researcher, news provider, mediator, government information, and campaign aide. Each role card defines expected behavior over six dimensions: user-agency fidelity, epistemic integrity, viewpoint symmetry, curation accountability, deliberative standing, and non-manipulation/refusal integrity.

Mirrored civic prompt pairs. Some prompts are mirrored pairs. The intended difference is viewpoint, not legality or task type. One prompt asks for one side of a civic argument or policy framing; the paired prompt asks for the counterpart framing. If the same model-role-mode setting refuses one side and answers the other, that is a concrete viewpoint-symmetry failure.

Judging. The primary judge is `xai:grok-4.3`. A stratified 300-row baseline sample is also judged by `qwen3:8b`. The two judges agree on baseline refusal labels for 81.7% of sampled rows, and the post-stratified refusal mismatch rate is 5.2%. Role-profile scores are less stable than refusal labels, so role-fit claims should be read as LLM-judge-calibrated until human labels are imported.

3 Where the Numbers Come From

The main baseline count is a full factorial grid:

$$5 \text{ models} \times 7 \text{ roles} \times 30 \text{ prompts} \times 2 \text{ role-presentation modes} = 2100 \text{ rows}.$$

Each row is generated once and then scored once by the primary judge. The raw generations are stored under `runs/adfe_v2_clean_local_grok/generations.jsonl`. The primary judge scores are stored under `runs/adfe_v2_clean_local_grok/v2/xai_grok-4.3/scores.jsonl`. The aggregate analysis is stored in `runs/adfe_v2_clean_local_grok/v2/analysis.json`. Paper tables and macros are regenerated from those artifacts by `adfe_runner build-paper-artifacts`; the paper does not hand-enter the headline counts.

The 420 mirrored-pair comparisons come from six matched prompt pairs. For each matched pair, the comparison is repeated under the same five models, seven roles, and two role-presentation modes:

$$6 \text{ prompt pairs} \times 5 \text{ models} \times 7 \text{ roles} \times 2 \text{ modes} = 420 \text{ pair comparisons.}$$

For each pair comparison, the analysis holds `model`, `role`, `agency_mode`, and `cycle` fixed and compares only the two matched prompt IDs.

4 Empirical Definitions

We use the following operational definitions. They are intentionally behavioral rather than ideological.

Context effect. A context effect is any change in model behavior associated with role, topic, prompt framing, or role-presentation mode under otherwise matched conditions. Context effects are expected in this setting; they are not automatically bias.

Fairness failure or behavioral bias. The primary fairness signal is unjustified asymmetric treatment of comparable civic requests. The cleanest measured case is one-sided refusal in a matched prompt pair: for the same model, role, role-presentation mode, and cycle, one prompt is refused and its counterpart is answered. Formally, for matched pair members a and b, one-sided refusal is

$$|\mathbb{1}[\text{refusal}_a] - \mathbb{1}[\text{refusal}_b]| = 1.$$

The one-sided refusal rate is the mean of this indicator over all matched-pair comparisons. This is why 72 / 420 is the central baseline fairness statistic.

Opinionatedness. This experiment does not estimate a left-right ideology score or a party-preference latent variable. It measures opinionated behavior indirectly through refusal asymmetry, viewpoint-symmetry scores, and role-inappropriate persuasion. A separate opinion-valence study would be needed to say that a model consistently prefers one political position.

Usefulness. Usefulness is measured as access plus answer quality. Access is whether the model answers instead of refusing. Answer quality is scored only among non-refusals on six 0–1 dimensions (U, E, V, C, D, M): user-agency fidelity, epistemic integrity, viewpoint symmetry, curation accountability, deliberative standing, and non-manipulation/refusal integrity. Refusals are not folded into the quality mean; refusal is measured separately.

Role fit. Role fit asks whether the output matches the assigned civic role. The judge assigns six 0–1 role-profile scores, and the reported role-profile fit is their mean. This is different from answer quality: an answer can be fluent and informative while still sounding like the wrong role.

5 Metrics

The evaluation separates outcomes that are often collapsed into one score.

Table 1: Baseline headline metrics generated from run artifacts.

Metric	Value
Primary judged outputs	2100
Non-refusal outputs used for quality	1800
Refusal rate	14.3%
Over-refusal rate	13.1%
One-sided mirrored-pair refusals	72 / 420
Alternate-judge sample rows	300
Alternate-judge refusal agreement	81.7%

Refusal and over-refusal. We measure the refusal rate, the over-refusal rate, topic-level refusal, and one-sided refusal across mirrored prompt pairs. A refusal is counted as over-refusal when the judge marks it unwarranted for the in-scope civic request.

Quality among answers. Quality is averaged only among non-refused outputs. This avoids hiding refusal behavior inside a single quality score.

Role-profile fit. Role-profile fit asks whether the output matches the assigned civic role. For example, a government-information response should emphasize procedural clarity and non-persuasion; a campaign aide may help with persuasion in role-appropriate ways but should not deceive or manipulate.

Regression gate. The release gate is built from the worst baseline failures. A candidate role policy fails the gate if it increases one-sided refusal, worsens over-refusal, or worsens role-profile fit on those gated rows.

Matched deltas and uncertainty. For remediation, each candidate row is matched to its baseline counterpart by (`model`, `role`, `agency_mode`, `prompt_id`). For binary metrics such as refusal, the per-row delta is -1, 0, or $+1$. For score metrics, the per-row delta is the candidate score minus the baseline score. We report the mean paired delta and an approximate 95% confidence interval computed as $\bar{x} \pm 1.96\, s/\sqrt{n}$ over the paired deltas.

6 Baseline Results

The baseline shows that role assignment is a real source of civic behavior variation. The primary judge marks 14.3% of rows as refusals and 13.1% as over-refusals. Across mirrored viewpoint comparisons, 72 of 420 pairs have one-sided refusal (17.1%).

The main baseline takeaway is not a single ideological direction. The actionable failure is instability: the same civic task can be answered or refused depending on role framing, mirrored prompt pairs can receive asymmetric treatment, and fluent answers can miss the assigned role's institutional duty.

Table 2: Baseline versus role-policy remediation. Deltas are candidate minus baseline.

Metric	Baseline	Remediation	Matched delta
Refusal rate	14.3%	13.4%	-0.9 pp
Over-refusal rate	13.1%	12.4%	-0.9 pp
One-sided mirrored-pair refusal	17.1%	8.8%	–
Role-profile fit	–	–	-0.01
Non-refusal quality	–	–	-0.031

7 Role-Policy Remediation

The remediation experiment asks a simple engineering question: if role prompts are policy controls, can a clearer executable role policy repair the failures? The remediation arm keeps the models, prompts, source packets, roles, role-presentation modes, generation settings, and primary judge fixed. The intended intervention is only the role-policy addendum: allowed help, forbidden persuasion, refusal criteria, source requirements, uncertainty language, escalation triggers, and a viewpoint-parity rule.

On the full matched grid, remediation modestly reduces aggregate refusal and over-refusal. Refusal falls from 14.3% to 13.4% (-0.9 pp matched delta; approximate 95% CI [-2.3 pp, +0.6 pp]), and over-refusal falls from 13.1% to 12.4% (-0.9 pp matched delta; approximate 95% CI [-2.2 pp, +0.5 pp]). The more important result is on mirrored pairs: one-sided refusals fall from 72 to 37 (17.1% to 8.8%). This is the clearest evidence that the policy addendum changed a concrete civic failure mode.

The intervention is not free. Matched role-profile fit changes by -0.01, and matched non-refusal quality changes by -0.031 (approximate 95% CI [-0.04, -0.022]). In other words, the policy makes the system less likely to refuse one side of a matched pair, but the answers it newly allows are not automatically better by every judged dimension.

8 Targeted Failures, Ablations, and Stress Tests

The full grid is dominated by ordinary rows where both policies behave similarly. To test whether the intervention repairs known failures, we also evaluate a targeted 300-key sample: refusal-asymmetry examples, role-profile misses, judge-disagreement examples, and low-disagreement controls. On this failure-heavy sample, refusal falls from 81.0% to 51.0% (-30.0 pp), over-refusal falls from 77.0% to 49.7%, role-profile fit improves by +0.17, and non-refusal quality improves by +0.045. This result is stronger than the full-grid average because the sample intentionally concentrates on the rows the baseline got wrong.

The regression gate gives the same story in release-monitoring form. On gated failures, one-sided refusal falls from 80.0% to 21.1% (-58.9 pp), over-refusal changes by -35.8 pp, and role-profile fit changes by +0.235. The candidate passes the gate.

The ablation study tests four removals from the policy addendum: no explicit viewpoint-parity rule, no refusal criteria, no source/uncertainty language, and no role-specific rules. All four ablated arms still improve over the baseline on the targeted sample, which means the global policy wrapper does real work. The most damaging removal is the role-specific rules: it gives back the most refusal and role-fit improvement relative to the full policy. The explicit viewpoint-parity line is not independently decisive in this sample, likely because other global instructions already imply viewpoint-neutral treatment of comparable in-scope requests.

The stress set contains 840 explicit-mode rows built from harder mirrored civic prompts. Stress refusal changes from 12.1% to 11.4%, and one-sided stress-pair refusal changes from 10.5% to 8.6%. The stress result supports a more cautious claim: remediation helps the most obvious failures, but it does not uniformly improve every role, topic, or model.

9 Interpretation: Context, Fairness, and Bias

The completed experiments support five claims.

First, context changes model behavior. Assigned role, topic, and prompt framing all affect refusal, role fit, and viewpoint-symmetry behavior under otherwise matched conditions. This means civic bias cannot be summarized only as a model-level left-right preference. It also appears as context-conditioned access to assistance.

Second, not every context effect is unfair. Some role differences are expected and justified. A government-information service should avoid persuasion; a campaign aide can provide persuasive help within non-deceptive limits; a mediator should map disagreements rather than win an argument. A role-conditioned difference becomes a fairness problem when the role does not justify the change: for example, when one side of a matched civic pair is refused and the counterpart is answered, or when a role silently changes evidentiary burden across political positions.

This pattern is plausible but only partly justifiable. It makes sense that contested civic topics and institution-like roles trigger more caution. It also makes sense that role-specific policy rules matter more than a generic instruction to be fair. What does not make sense as a product behavior is unexplained asymmetry within matched prompt pairs. That is the fairness signal this evaluation is designed to isolate.

Third, the most actionable baseline failure is asymmetric civic service. The issue is not simply that the model has one ideological direction. It is that comparable civic requests sometimes receive different refusal treatment, which changes who can obtain usable assistance.

Fourth, clearer role policy can repair important failures, especially on known bad cases and mirrored-pair release gates. The intervention roughly halves one-sided refusals on the full grid and substantially improves the targeted failure sample.

Fifth, remediation has tradeoffs. Some roles and models improve more than others, and non-refusal quality slightly declines on the full matched grid. A deployer should therefore treat role policy as a versioned control surface that needs repeated evaluation, not as a one-time prompt fix.

10 Related Work

This work follows holistic evaluation in separating capabilities and harms rather than reporting a single score [2]. It is related to social bias and opinion benchmarks such as BBQ and OpinionQA [5, 9], and to political-preference evaluation of language models [7, 8]. The difference is that we vary assigned deployment role as the experimental variable rather than treating the system as one role-less respondent.

Persona-prompting work shows that role prompts can reshape outputs in conditional and unreliable ways [12, 4, 10]. We build on that observation by treating civic roles as policy-bearing configuration. XSTest motivates explicit over-refusal measurement [6], TruthfulQA motivates judge calibration for factuality [3], and MT-Bench plus LLM-as-judge surveys motivate scalable judging while warning that judge agreement is itself an empirical question [11, 1].

11 Limitations

The current evidence is limited to small local models, U.S. civic topics, a static prompt set, and LLM judges. The exploratory frontier arm contains 630 rows and a 5.7% refusal rate, but it is not pooled with the local evidence because same-provider generation and judging make it exploratory rather than independent. The alternate-judge results show that refusal labels are more robust than role-profile scores. Human review has been designed and exported, but completed two-rater labels are not yet imported into the results reported here. The current claims therefore should be read as LLM-judge-calibrated findings, with human validation still needed for a submission-ready version.

12 Conclusion

Role prompts are a control surface for civic AI policy. Treating them as mere style text hides failures that matter in deployment: one-sided refusal of comparable civic requests, role-inappropriate help, unstable judge conclusions, and uneven remediation effects. A role-counterfactual evaluation makes these failures measurable and gives system builders a concrete loop: version role policies, test mirrored viewpoints, compare judges, route hard cases to human review, and block releases when role changes worsen civic obligations.

References

[1] Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. LLMs-as-judges: A comprehensive survey on LLM-based evaluation methods, 2024.

[2] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

[3] Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of ACL 2022*, 2022.

[4] Marlene Lutz, Indira Sen, Georg Ahnert, Elisa Rogers, and Markus Strohmaier. The prompt makes the person(a): A systematic evaluation of sociodemographic persona prompting for large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025.

[5] Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. BBQ: A hand-built bias benchmark for question answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022.

[6] Paul Rottger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of NAACL 2024*, 2024.

[7] David Rozado. The political preferences of LLMs. *PLOS ONE*, 2024.

[8] David Rozado. Measuring political preferences in ai systems: An integrative approach, 2025.

[9] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect?, 2023.

[10] Shuai Xiao, Su Liu, Weikai Zhou, Jialun Wu, Xinjie He, Zhiyuan Lin, and Qiyang Xie. When does persona prompting actually help? a retrieval and metric analysis of expert role injection in LLMs, 2026.

[11] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena, 2023.

[12] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. When "a helpful assistant" is not really helpful: Personas in system prompts do not improve performances of large language models, 2023.

A Artifact Notes

All headline numbers are read from `generated/numbers.tex`, which is produced by `adfe_runner build-paper-artifacts`. Tables under `generated/tables/` are regenerated from run artifacts and should be treated as the source of truth.

B Example Prompt Families

The prompt set includes civic briefings, government-information requests, mediation tasks, and mirrored argument requests. Example task shapes include: explain what people should know about a policy area using the supplied source packet; summarize procedural public information without persuasion; map tradeoffs between two sides; and draft an argument for one side of a policy debate with a mirrored counterpart for the other side.